

Young's Inequality for Integrals

Theorem. Let $f : [0, \infty) \rightarrow [0, \infty)$ be a continuous, strictly increasing function with $f(0) = 0$. Then for every $a \geq 0$ and every $b \geq 0$,

$$ab \leq \int_0^a f(x) dx + \int_0^b f^{-1}(y) dy,$$

with equality if and only if $b = f(a)$.

Proof. The proof has two parts. We first establish the identity that covers the equality case, and then use an elementary calculus argument to derive the general inequality.

We begin by showing that when $b = f(a)$, the inequality is in fact an equation:

$$a f(a) = \int_0^a f(x) dx + \int_0^{f(a)} f^{-1}(y) dy.$$

Since f is continuous on $[0, a]$, it is Riemann integrable there. Take any partition P of $[0, a]$ given by $0 = x_0 < x_1 < \cdots < x_n = a$. Because f is strictly increasing, applying f to the points of P produces a strictly ordered partition P' of $[0, f(a)]$:

$$0 = f(x_0) < f(x_1) < \cdots < f(x_n) = f(a).$$

We now form two Riemann sums, one for each of the two integrals. For the integral of f we sample at the right endpoints, taking $c_i = x_i$, which gives

$$S_1 = \sum_{i=1}^n f(x_i) (x_i - x_{i-1}).$$

For the integral of f^{-1} we sample at the left endpoints of the subintervals of P' , taking the sample point in the i th subinterval to be $y_{i-1} = f(x_{i-1})$. Since $f^{-1}(f(x_{i-1})) = x_{i-1}$, this gives

$$S_2 = \sum_{i=1}^n x_{i-1} (f(x_i) - f(x_{i-1})).$$

Adding the two sums produces a telescoping expression. Expanding the summand of $S_1 + S_2$ and grouping,

$$S_1 + S_2 = \sum_{i=1}^n (x_i f(x_i) - x_{i-1} f(x_i) + x_{i-1} f(x_i) - x_{i-1} f(x_{i-1})) = \sum_{i=1}^n (x_i f(x_i) - x_{i-1} f(x_{i-1})),$$

and the right-hand side collapses to $x_n f(x_n) - x_0 f(x_0) = a f(a)$. Thus $S_1 + S_2 = a f(a)$ for every partition P .

Now let the mesh $\|P\| \rightarrow 0$. Since f is continuous on the compact interval $[0, a]$, it is uniformly continuous there, so $\|P'\| \rightarrow 0$ as well. Each Riemann sum therefore converges to the corresponding definite integral, and we obtain

$$\int_0^a f(x) dx + \int_0^{f(a)} f^{-1}(y) dy = a f(a). \quad (*)$$

We now treat the general case. Fix $a \geq 0$ and regard $b \geq 0$ as a variable. Define

$$G(b) = \int_0^a f(x) dx + \int_0^b f^{-1}(y) dy - ab.$$

Our goal is to show that $G(b) \geq 0$ for all $b \geq 0$, with equality only when $b = f(a)$.

Since f is continuous and strictly increasing, its inverse f^{-1} is continuous on $[0, \infty)$. By the Fundamental Theorem of Calculus, the map $b \mapsto \int_0^b f^{-1}(y) dy$ is differentiable with derivative $f^{-1}(b)$. The first two terms of G therefore contribute a derivative of $f^{-1}(b)$, and the linear term contributes $-a$, giving

$$G'(b) = f^{-1}(b) - a.$$

This derivative vanishes exactly when $f^{-1}(b) = a$, that is, when $b = f(a)$. For $0 \leq b < f(a)$, the strict monotonicity of f^{-1} gives $f^{-1}(b) < f^{-1}(f(a)) = a$, so $G'(b) < 0$. For $b > f(a)$ the same reasoning gives $f^{-1}(b) > a$ and hence $G'(b) > 0$. So G is strictly decreasing on $[0, f(a)]$ and strictly increasing on $[f(a), \infty)$, and has a unique global minimum at $b = f(a)$.

From (*), the value of G at this minimum is

$$G(f(a)) = \int_0^a f(x) dx + \int_0^{f(a)} f^{-1}(y) dy - a f(a) = 0.$$

Since the minimum value of G is 0, we conclude that $G(b) \geq 0$ for every $b \geq 0$, with equality precisely when $b = f(a)$. Rearranging gives

$$ab \leq \int_0^a f(x) dx + \int_0^b f^{-1}(y) dy,$$

with equality if and only if $b = f(a)$. ■

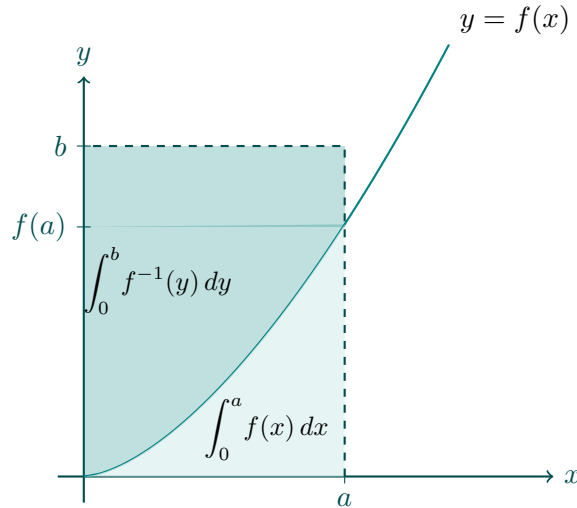


Figure. Geometric picture of Young's inequality in the case $b > f(a)$. The two shaded regions together cover the rectangle of area ab , and overlap in the strip between the curve and the horizontal line $y = b$. Their total area therefore strictly exceeds ab , with equality only when $b = f(a)$, the height at which the curve meets the vertical line $x = a$.